ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

Final Project Report

SWE 4304: SPL-I

Project Title

School Admission System

Team Members

Talha Zubayer (230042119)
Sheikh Irtiaj Ahmed (230042130)
Md Nadid Hassan Khan (230042146)

Supervised By

Faisal Hussain
Assistant Professor
Department of Computer Science and Engineering

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Problem Statement

In Bangladesh, the school admission process in many cases is still affected by manual paperwork, long queues, lack of centralized verification, and limited transparency in seat allocation. Students and guardians often face difficulties while applying to multiple schools, verifying personal information, checking eligibility, and understanding admission results. Manual systems can also lead to errors, duplicate applications, misuse of information, and confusion regarding quota-based admission and lottery results.

Another major problem is that guardians have to provide the same information repeatedly in different places, while school authorities also spend significant time checking applicant data manually. In urban areas especially, where competition for admission is high, the absence of a secure and automated admission process can create inefficiency, delays, and questions about fairness.

Therefore, there is a need for a secure, automated, and transparent school admission system in the context of Bangladesh that can verify applicant information, manage school choices, support quota-based applications, and conduct lottery-based seat allocation fairly.

Potential Solution

To overcome the challenges of manual and inefficient school admission processes, a fully digitalized admission system can be introduced. Such a system would centralize applicant data, automate verification processes, and reduce the need for repetitive form submissions. By integrating secure identity verification (such as birth certificate and guardian NID validation), OTP-based confirmation, and automated eligibility checking, the system can ensure accuracy and authenticity of information. Additionally, implementing a transparent, rule-based lottery mechanism for seat allocation, along with real-time updates and notifications, can significantly improve fairness and reduce confusion among applicants and guardians. A centralized digital platform would also minimize paperwork, shorten processing time, and enhance overall accessibility for users across different regions.

Our project addresses these challenges by implementing a School Admission System tailored to the Bangladeshi context. It digitalizes key admission steps including applicant registration, birth certificate and NID verification, OTP-based mobile confirmation, eligibility checking, and structured school, shift, and quota selection. The system ensures transparency through a lottery-based seat allocation process aligned with available seats and quota policies. It also reduces redundancy by allowing applicants to

manage their information in a single platform and provides features like applicant ID recovery and result checking. For administrators, the system simplifies school management and automates the admission workflow, thereby improving efficiency and reducing manual errors. Overall, the project effectively applies digital solutions to make the admission process more organized, reliable, and fair.

Implemented Features

Application Portal

- Student can return to main menu anytime.
- Encapsulated choice information in number as much as possible to minimise user error
- Case independent scanning throughout the project
- Apply for a new form
    - Automatically fetches Student's name, Parent's name, date of birth from the birth certificate number [minimising trivial error from user]
    - If a student has one application and again applying for another, it will auto-complete the similar fields of another application
    - Verify mobile number to Valid Phone number Syntax
    - Verify ownership of Mobile number through an OTP
    - Auto-completes guardian's information from NID (minimising trivial error)
    - Checks if the NID given for parents are actually the student's parents or not.
    - In case of absence of parents, mandating the information for local guardian
    - Checks for age eligibility to be admitted into certain class
    - Narrowing down users address selection step by step [to minimise trivial error]
    - Automatically shows all available schools in a certain thana, after choosing the thana.
    - Restricts student from choosing same seating in multiple choice of an application
    - Restricts same student to choose same seating in a different application.
    - Let's student apply for a quota in each choice separately
    - Verifies various quota with the appropriate reference ID (for FF), automatically matching information (for CA)
    - Creates an Application Summery before confirming

  - Let's user re edit any field accordingly, and handles further inconsistency after that (e.g. cancelling school choices if present address changes and CA quota nullifies)
  - After confirming the choices, the application is saved in to database and in a txt form in a separate file.

- Vacant Seat
  - User can select a thana from choosing division, then district and can see all the available schools in that thana.
  - User can choose a school, school's class to see available blank seat (grouped by shift) in that school's class.
- Recover/Delete Applicant ID
  - Verifies ownership of application through OTP verification. [Phone number is hidden]
- Show Result
  - Shows the result only when lottery is done.
  - Shows admitted school, class, shift of a student based on birth certificate number.

School Authority

- Login Portal for individual school's profile
- Login with Unique EIIN and Password [Hidden in terminal and hashed in database]
- Authority can update school's status by creating, updating OR delete seating.
- After result is published, authority can see all admitted students of his school with their parents contact number grouped by classes.

Master Admin

- Login portal for Admin
- Login with Admin ID and Password [Hidden in terminal and hashed in database]
- Create school with specifying name, and postcode of the school. Automatically generating a random EIIN and password for School authority to log into
- Can delete school
- Can implement the Admission Lottery ()

Tools and technology

Programming Language: The entire project is developed using **Java**.

IDE: We used **Visual Studio Code** to write, test, and organize our code efficiently.

Version Control: For collaboration and tracking changes, we used **Git and GitHub**.

Git (Local Control)

GitHub (Collaboration on Cloud)

Data Storage: Since databases were restricted for this project, we implemented a **file-based** storage system to save and manage all necessary information.

Application Domain

- Public school/college admission lottery & seat allocation
- University hostel room allocation
- Scholarship selection with eligibility + quota
- Government training allocation (limited slots, district/area rules)
- Job allocation with category quotas and verification (NID/birth certificate)

Project Timeline

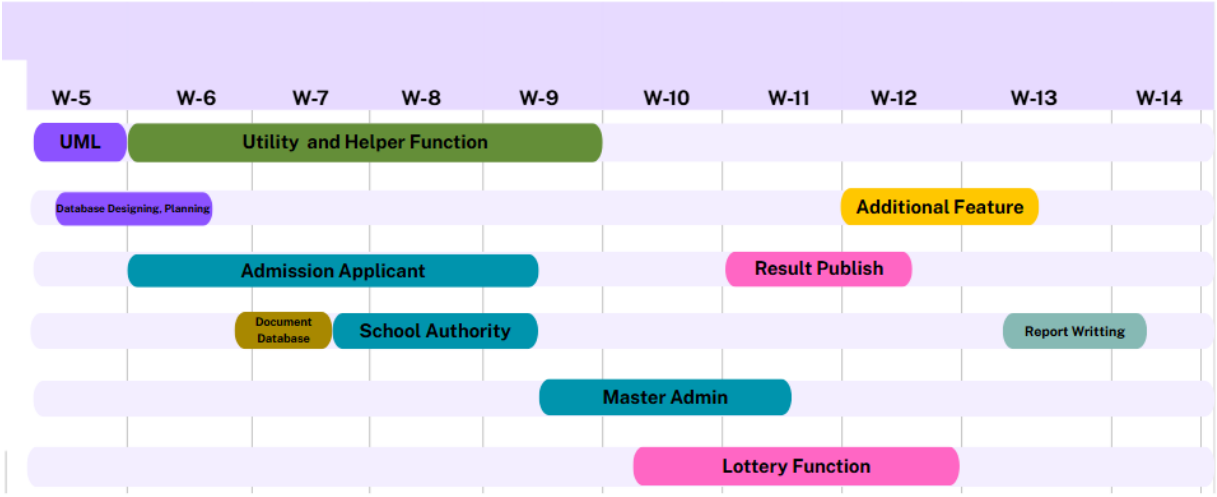
Week-5: Our team focused on designing the complete UML diagram. By the middle of that week, we also began database design.

Week-6: The database design was completed by mid-week. At the same time, we started developing utility and helper functions, and also began working on the Admission Applicants module. By the end of that week, we injected initial information into the database.

Week-7 to Week-9: We continued developing the Admission Applicants module, which was completed by mid-week 9. Alongside this, we worked on the School Authority module. Both modules are expected to be finished before our Progression Presentation.

Mid-Week 9 to the end of Week 11: We developed the Master Admin module and one of the core features, the Lottery function, which took several weeks to complete.

Week-12: We implemented Result Publishing based on the lottery outcomes, added additional features, and completed the final project report.



Contribution

Talha Zubayer (ID: 230042119)

- School Admission Portal
- New Applicant
- Recover/Delete Application
- See Vacant Seat
- Primary Database Designing and Class Diagrams
- Primary Project workflow

Sheikh Irtiaj Ahmed (ID: 230042130)

- School Authority Portal (Entirely)
- Add new class, Show Particular Schools info, Update Seat allocation
- Print Result in both School Authority and Student Admission Portal
- Designing and Creating additional Databases for Login mechanism and Database Encrypting

Md Nadid Hasan Khan (ID: 230042146)

- Implementing SQL like emulation through java library functions on File Database
- Populating All databases and maintaining integrity constraint
- Master Admin Portal (Entirely)
- Designing and Creating additional Databases for Lottery and it's result.

Future work

- Migrate from File Database system to SQL based database.
- Extend for supporting multiple quota in a single choice dynamically
- Option for running multiple lottery and Migration upon choice confirmation from student
- Proper notification system through SMS and Email
- UI\UX Dashboard implementation
- Implementing on live server on web portal
- Implementing API usage
- Instead of text form, actually creating and recovering organised pdf style form
- Working with a more populated, repository made Databases
- Stronger Security and Verification
- Implementing Payment Gateway

Source Code

https://github.com/nadid-hk/SPL-1-23